

A Hybrid Evolutionary Algorithm for the Multi-Objective Capacitated Lot-Sizing Problem

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Abstract—Existing studies on the capacitated lot-sizing problem (CLSP) often aggregate multiple costs into a single weighted objective. However, real-world production planning usually involves multiple conflicting objectives and requires balancing trade-offs among competing goals. While multi-objective extensions of CLSP (MO-CLSP) have attracted considerable attention in both research and practice, publicly available benchmark instances for MO-CLSP remain limited. To enable fair and reproducible comparisons, we construct a benchmark set by adapting standard single-objective CLSP instances to the MO-CLSP. In the benchmark, we consider three objectives, holding cost, setup cost, and overtime cost, which are widely considered in classical lot-sizing models and practical production planning. We further propose a multi-objective evolutionary algorithm (MOEA) for the MO-CLSP that incorporates problem-specific strategies and a cluster-guided large neighborhood search to improve convergence and diversity. Comparative experiments with widely used MOEAs show that the proposed method is competitive and often provides better Pareto front approximations. Further analyses are also included to validate the effectiveness of the algorithm components.

Index Terms—Capacitated lot-sizing, Benchmark, Multi-objective evolutionary algorithm, Multi-objective optimization

I. INTRODUCTION

Lot-sizing is a classical and fundamental production planning problem that determines production quantities and inventory levels over a finite planning horizon. Among its variants, the capacitated lot-sizing problem (CLSP) incorporates capacity limitations and has been widely studied in the literature [1]. Beyond theoretical interest, the CLSP and its extensions have also been successfully used in many real-world applications to support planning decisions involving production, inventory, and capacity usage, such as papermaking [2], glass container production [3], and cloud manufacturing [4], etc.

Traditionally, the CLSP is formulated as a single-objective optimization problem by aggregating multiple cost components into a weighted objective. These cost components typically reflect different operational aspects of production planning, including inventory holding, setup operations, overtime (or capacity expansion), and backordering (or lost sales). In practice, these aspects are often managed by different departments with different priorities, and decision makers may need to balance conflicting objectives rather than rely on a single integrated criterion. Consequently, their relative importance must be specified in advance by choosing weights, which can be difficult to determine in practice. Moreover, the trade-offs implied by these weights may change with managerial preferences, departmental responsibilities, or operating conditions. As a result, solutions can be sensitive to the chosen weights, which means that a fixed set of weights reflects one preference among decision makers and reveals only a limited range of trade-offs.

This has led to increasing interest in multi-objective CLSP (MO-CLSP) models, which consider multiple cost components as separate objectives and support decision makers in exploring alternative trade-offs. Such MO-CLSP formulations can be addressed within the general framework of multi-objective optimization, aiming to approximate the Pareto front (PF) that reflects the trade-offs among objectives. To approximate the PF effectively, one common choice is multi-objective evolutionary algorithms (MOEAs), which can generate a set of well-distributed trade-off solutions in a single run. In this context, several studies have investigated extensions and variants of MO-CLSP under different modeling assumptions and application settings, where the objectives are often defined according to the targeted extension. For example, MO-CLSP

extensions may incorporate supplier selection with objectives such as cost, quality, and service level [5] or cost and carbon emissions [6], and may also include objectives such as storage-space usage and workload balance [7].

Although multi-objective variants of the CLSP have been studied, most existing works focus on specific problem variants and are evaluated on case-specific or randomly generated instances. To the best of our knowledge, the instances and implementations used in these studies are not always publicly released. Consequently, publicly available benchmark sets for the MO-CLSP remain limited, making it difficult to validate, extend, and compare algorithms under consistent instances and settings.

In this paper, instead of focusing on problem-specific extensions, we study a baseline MO-CLSP that preserves the standard CLSP structure and considers three cost objectives (inventory holding cost, setup cost, and overtime cost) that are widely used in lot-sizing models and practical production planning. To solve this problem, a hybrid MOEA (MOEA-CLSP) is proposed by integrating a cluster-guided large neighborhood search (LNS) to enhance convergence while maintaining a well-distributed set of solutions. A benchmark set is constructed by adapting widely used single-objective CLSP instances, and the benchmark data, source code, and detailed algorithm parameter settings are released to support reproducible comparisons.

II. PROBLEM DESCRIPTION

The CLSP can be viewed as a capacity-constrained extension of the Wagner-Whitin problem, which was introduced in the 1950s [8], [9]. As a widely studied standard variant of the lot-sizing problem, the CLSP is often used as a baseline model in the literature. It determines, over a finite multi-period horizon, when and how much to produce for each item and how much inventory to hold, under period-specific demand, capacity limits, and setups.

Fig. 1 gives an illustrative example of the CLSP over several periods, where the horizontal axis represents the planning periods. The upper panel focuses on capacity utilization. In each period, total capacity utilization is composed of the usage caused by producing different items and the setup usage incurred when an item is produced in that period. The blue line indicates the capacity limit, which may vary across periods. If the total utilization exceeds this limit, the excess can be covered by overtime at an additional cost.

The lower panel illustrates the corresponding inventory flow. In each period, the available supply for each item consists of inventory carried over from the previous period and production in the current period. This supply is used to satisfy demand for each item, and the inventory held at the end of the period is carried over to the next period, typically incurring holding cost. Consequently, decisions are coupled over time through inventory carry-over and across items through competition for shared capacity within each period.

Overall, the CLSP captures a fundamental trade-off among capacity utilization, setups, and inventory. Producing earlier

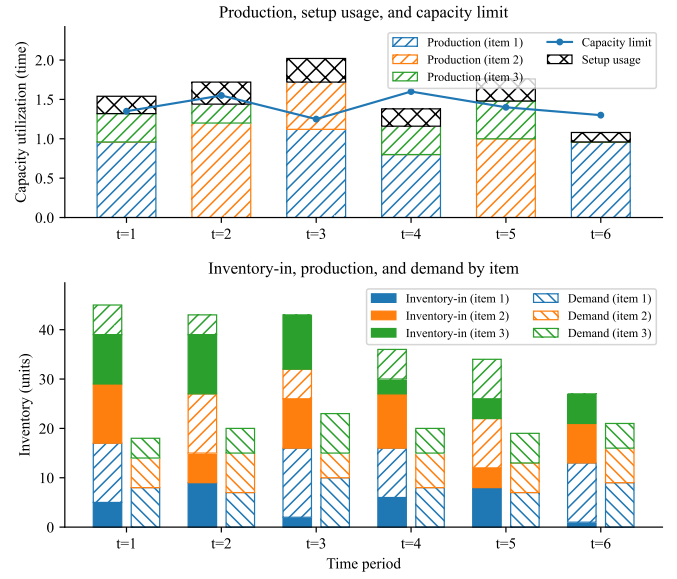


Fig. 1. Illustration of the CLSP over a finite horizon.

can smooth capacity utilization and reduce future setups, but it increases inventory and thus holding cost. Producing later can reduce inventory and holding cost, but it may require more frequent setups and tighter capacity allocation, which typically increases setup cost and overtime cost. Accordingly, the objective in this study is to determine production and inventory decisions that satisfy demand while simultaneously minimizing holding cost, setup cost, and overtime cost.

III. MATHEMATICAL MODEL

The proposed formulation is developed with reference to standard CLSP models in the literature [10]–[13] and adapted to consider multiple objectives, including inventory holding cost, setup cost, and overtime cost.

A. Sets and indices

J items $j \in \{1, \dots, |J|\}$
 T periods $t \in \{1, \dots, |T|\}$

B. Parameters

C_t regular capacity in period t
 OC_t cost of one unit of overtime in period t
 A_j capacity consumption per unit of item j
 ST_j capacity consumption caused by setting up item j
 SC_j setup cost for item j
 H_j holding cost for one unit of item j per period
 D_{jt} demand of item j in period t
 i_{j0} initial inventory of item j at the beginning of the planning horizon

C. Decision variables

$x_{jt} \geq 0$ production quantity of item j in period t
 $i_{jt} \geq 0$ inventory of item j at the end of period t
 $y_{jt} \in \{0, 1\}$ setup indicator for item j in period t
 $o_t \geq 0$ overtime used in period t

D. Model Equations

$$\min F_1 = \sum_{j \in J} \sum_{t \in T} H_j i_{jt} \quad (1)$$

$$\min F_2 = \sum_{j \in J} \sum_{t \in T} SC_j y_{jt} \quad (2)$$

$$\min F_3 = \sum_{t \in T} OC_t o_t \quad (3)$$

$$i_{j,t-1} + x_{jt} = D_{jt} + i_{jt} \quad \forall j \in J, t \in T \quad (4)$$

$$\sum_{j \in J} (A_j x_{jt} + ST_j y_{jt}) \leq C_t + o_t \quad \forall t \in T \quad (5)$$

$$x_{jt} \leq \left(\sum_{\tau \in T} D_{j\tau} \right) y_{jt} \quad \forall j \in J, t \in T \quad (6)$$

$$x_{jt}, i_{jt} \geq 0; y_{jt} \in \{0, 1\} \quad \forall j \in J, t \in T \quad (7)$$

$$o_t \geq 0 \quad \forall t \in T$$

Equations (1), (2), and (3) minimize holding, setup, and overtime costs, respectively. Constraint (4) defines the inventory balance between consecutive periods. Constraint (5) limits the regular capacity while allowing overtime usage. Constraint (6) links production decisions to setup decisions, and (7) defines the variable domains.

IV. SOLUTION APPROACH

To solve the MO-CLSP, we propose a hybrid evolutionary algorithm that incorporates problem-specific strategies and embeds a cluster-guided LNS. The evolutionary component ensures population diversity and global exploration, while the LNS component further improves selected solutions in different regions of the objective space.

A. Solution Representation

A solution is encoded as a real-valued production matrix

$$\mathbf{X} = (x_{jt}) \in \mathbb{R}_{\geq 0}^{|J| \times |T|}, \quad (8)$$

where each entry x_{jt} corresponds to the production quantity of item j in period t in the MO-CLSP model. This encoding directly represents the primary decision variable of the model.

Given a production matrix $\mathbf{X}$, the remaining variables can be derived during evaluation. Inventory levels i_{jt} are uniquely determined by the inventory balance constraint (4). Setup variables y_{jt} are defined by production activity, with $y_{jt} = 1$ if $x_{jt} > 0$ and $y_{jt} = 0$ otherwise. Overtime usage o_t is computed whenever the regular capacity constraint (5) is violated.

Consequently, the evolutionary search is conducted only over the production quantities in $\mathbf{X}$. Demand satisfaction and capacity constraints are handled during evaluation, with repairs applied when necessary. This design streamlines the representation and focuses the search on the primary planning decisions.

Algorithm 1: MOEACLSP

Input: population size N ; number of clusters k ; LNS iterations I ; relax ratio ρ ;

Output: final population P ;

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1  $P \leftarrow \text{Initialization}(N)$ ;
2 while termination condition is not satisfied do
3    $\text{parents} \leftarrow \text{Selection}(P, N)$ ;
4    $\text{offspring} \leftarrow \text{Recombination}(\text{parents})$ ;
5    $\text{offspring} \leftarrow \text{Production Shifting}(\text{offspring})$ ;
6    $\text{offspring} \leftarrow \text{Repair}(\text{offspring})$ ;
7    $Q \leftarrow [P, \text{offspring}]$ ;
8    $P \leftarrow \text{Environmental Selection}(Q, N)$ ;
9    $\text{reps} \leftarrow \text{Clustering}(P, k)$ ;
10  foreach  $x \in \text{reps}$  do
11     $x^{cur} \leftarrow x$ ;
12    for  $i \leftarrow 1$  to  $I$  do
13       $x' \leftarrow \text{LNS Step}(x^{cur}, \rho)$ ;
14      if  $x'$  dominates  $x^{cur}$  then
15         $x^{cur} \leftarrow x'$ ;
16    replace  $x$  in  $P$  by  $x^{cur}$ ;
17 return  $P$ ;

```

B. Algorithm Framework

Algorithm 1 presents the overall framework of the proposed hybrid evolutionary algorithm. The algorithm starts from an initial population P of size N . At each generation, offspring solutions are generated from the current population through recombination, followed by production shifting and repair operations. Environmental selection is then applied to the combined population based on dominance and crowding distance. In this dominance relation, any feasible solution is preferred to an infeasible one, and when both solutions are feasible, one solution dominates another if it is better in at least one objective and not worse in the others.

To further enhance search performance, a cluster-guided LNS is incorporated into the evolutionary process. For each representative solution selected from clusters in the objective space, LNS is applied for a fixed number of iterations, during which the solution may be updated based on dominance. After LNS, the updated solution is returned to the population. The above procedure is repeated until a termination condition is satisfied, and the final population is returned as an approximation of the Pareto-optimal solution set for the MO-CLSP.

C. Offspring Generation

Offspring solutions are generated from the current population through a sequence of recombination, production shifting, and repair operations. At each generation, parent solutions are randomly selected from the population and recombined to produce offspring solutions. The recombination operation integrates simulated binary crossover (SBX) and a structural mutation, where SBX performs real-valued recombination on production quantities. The structural mutation switches the

TABLE I
MEAN AND STANDARD DEVIATION OF HV (HIGHER IS BETTER) OBTAINED BY MOEACLSP AND OTHER ALGORITHMS.

Instance	MOEACLSP	MOEA1	MOEA2	MOEA3	MOEA4	MOEA/D	NSGA-II	NSGA-III
1	7.94e-01 _{7.20e-02}	7.43e-01 _{8.47e-02} =	2.90e-01 _{3.29e-02} +	4.17e-01 _{5.38e-02} +	4.22e-01 _{5.29e-02} +	1.68e-01 _{4.39e-02} +	2.47e-01 _{4.69e-02} +	2.15e-01 _{2.18e-02} +
2	8.24e-01 _{5.37e-02}	8.04e-01 _{4.91e-02} =	3.17e-01 _{3.44e-02} +	4.56e-01 _{4.62e-02} +	4.19e-01 _{4.93e-02} +	1.29e-01 _{3.37e-02} +	2.17e-01 _{4.39e-02} +	1.98e-01 _{3.84e-02} +
3	7.83e-01 _{3.95e-02}	7.04e-01 _{4.53e-02} +	4.13e-01 _{2.62e-02} +	5.63e-01 _{2.35e-02} +	4.17e-01 _{2.71e-02} +	3.57e-01 _{4.07e-02} +	4.39e-01 _{5.35e-02} +	4.10e-01 _{7.10e-02} +
4	7.81e-01 _{4.46e-02}	6.73e-01 _{3.70e-02} +	4.43e-01 _{2.83e-02} +	5.56e-01 _{3.29e-02} +	4.53e-01 _{4.52e-02} +	2.91e-01 _{4.49e-02} +	3.71e-01 _{3.29e-02} +	3.61e-01 _{2.97e-02} +
5	9.58e-01 _{7.39e-02}	8.04e-01 _{4.81e-02} +	7.54e-01 _{1.76e-02} +	7.15e-01 _{1.80e-02} +	7.09e-01 _{1.24e-02} +	7.47e-01 _{3.05e-02} +	7.12e-01 _{2.50e-02} +	6.89e-01 _{1.69e-02} +
6	9.81e-01 _{1.21e-01}	7.47e-01 _{3.79e-02} +	7.82e-01 _{2.25e-02} +	7.97e-01 _{5.68e-02} +	7.62e-01 _{1.39e-02} +	7.64e-01 _{3.25e-02} +	7.67e-01 _{4.73e-02} +	7.59e-01 _{2.86e-02} +
7	7.65e-01 _{4.37e-02}	5.30e-01 _{2.77e-02} +	5.89e-01 _{3.10e-02} +	5.37e-01 _{5.75e-02} +	4.25e-01 _{2.46e-02} +	1.96e-01 _{3.77e-02} +	3.81e-01 _{4.77e-02} +	3.25e-01 _{3.94e-02} +
8	7.65e-01 _{5.97e-02}	7.26e-01 _{5.54e-02} +	2.76e-01 _{3.05e-02} +	4.25e-01 _{5.32e-02} +	3.65e-01 _{5.21e-02} +	1.94e-01 _{2.24e-02} +	2.80e-01 _{4.33e-02} +	2.27e-01 _{2.79e-02} +
9	7.33e-01 _{4.90e-02}	6.90e-01 _{4.23e-02} +	3.21e-01 _{3.11e-02} +	4.62e-01 _{4.46e-02} +	3.63e-01 _{4.88e-02} +	2.25e-01 _{3.22e-02} +	2.68e-01 _{3.85e-02} +	2.42e-01 _{2.16e-02} +
10	7.08e-01 _{3.12e-02}	6.13e-01 _{4.20e-02} +	3.66e-01 _{3.33e-02} +	5.11e-01 _{5.52e-02} +	3.39e-01 _{3.99e-02} +	2.72e-01 _{3.19e-02} +	3.11e-01 _{3.95e-02} +	2.73e-01 _{2.13e-02} +
+/-/=		8/0/2	10/0/0	10/0/0	10/0/0	10/0/0	10/0/0	10/0/0

production activity independently for each item and each period with a given probability, enabling the exploration of alternative setup decisions and production structures.

To exploit the time-related structure of the MO-CLSP, a production shifting operator is subsequently applied to offspring solutions. This operator transfers part of the production quantity of an item from a later period to an earlier one, allowing adjustments of production timing. Such shifts introduce additional production plans into the search process.

Finally, a repair procedure is applied to each offspring solution to recover feasibility. During repair, inventory levels are updated over the planning horizon to track demand satisfaction. If inventory and production are insufficient, production is supplemented accordingly, potentially resulting in overtime.

D. Cluster-Guided LNS

To enhance local exploration, a cluster-guided LNS is embedded into the evolutionary framework. Rather than applying LNS to all individuals, the population is clustered in the objective space using k-means, and one representative solution per cluster is selected as the solution closest to the cluster center.

For each selected representative, the LNS procedure is independently applied for I iterations. At each iteration, a subset of periods with size determined by the relax ratio ρ is selected, and the corresponding production quantities are temporarily relaxed. The modified solution is then repaired and evaluated, and the current solution is updated if the modified solution dominates it. Through this cluster-guided application of LNS, the algorithm enables local exploration around representative solutions while preserving population diversity.

TABLE II
INSTANCE COMBINATIONS.

Index	1	2	3	4	5	6	7	8	9	10
T	15	15	15	15	20	20	20	30	30	30
J	6	8	12	24	10	20	30	6	12	24

V. COMPUTATIONAL EXPERIMENTS

A. Benchmark Instances

To enable reproducible comparisons on the MO-CLSP, we construct benchmark instances adapted from the classical

CLSP test set of Trigeiro, Thomas, and McClain (TTM) [12] ($T \in [15, 30]$, $J \in [6, 30]$)¹, which has been used by many studies and remains widely referenced [11], [13]–[16].

We consider 10 combinations with different planning horizons and numbers of items, covering all (T, J) pairs in the original TTM test set (Table II). For each combination, we include 5 instances adapted from the original TTM test set, resulting in 50 instances in total. All instances follow the classical multi-item, multi-period CLSP structure with deterministic demands, regular capacity limits, setup consumption, and overtime.

Due to space limitations, we report in the main text only one representative instance for each combination (i.e., 10 instances in total), while the complete results on all instances, together with the benchmark data, source code, and detailed parameter settings for each algorithm are made publicly available.²

B. Experimental Settings

MOEACLSP is compared with three popular MOEAs, NSGA-II [17], MOEA/D [18], and NSGA-III [19], together with several ablation variants. All algorithms use a population size $N = 100$ and are terminated after $5 \times J \times T \times N$ fitness evaluations, and each instance is independently solved 20 times. For each algorithm, its parameters are determined through orthogonal experiments, including operator probabilities, the number of clusters (k), the number of LNS iterations (I), the relax ratio (ρ), etc. For fairness, NSGA-II, MOEA/D, NSGA-III, and MOEACLSP share the same recombination and repair operators, which have been found to improve solution quality, while their algorithm-specific components remain unchanged. The contribution of these operators and other components is examined via an ablation study, where each variant removes one component from MOEACLSP while keeping the other settings fixed.

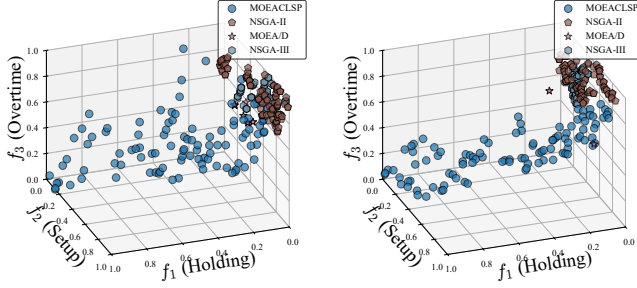
Solution quality is evaluated using HV [20] and IGD [21]. HV is computed with the reference point (1.1)³, and for each instance, IGD uses all feasible non-dominated solutions collected from all algorithms and runs as the reference set. The mean and standard deviation over 20 runs are reported, and the Wilcoxon rank-sum test at the 0.05 level is used to compare MOEACLSP with competitors. In the result tables,

¹<https://www.suerie.de/testsets.html>

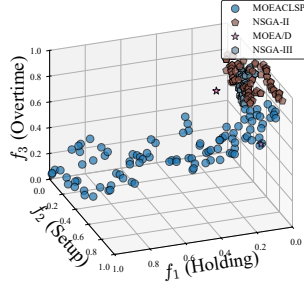
²<https://github.com/pomisula/moclsp>

TABLE III
MEAN AND STANDARD DEVIATION OF IGD (LOWER IS BETTER) OBTAINED BY MOEACLSP AND OTHER ALGORITHMS.

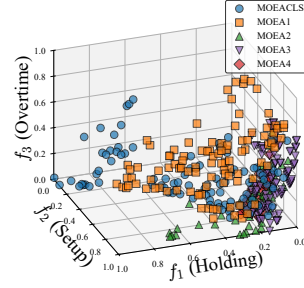
Instance	MOEACLSP	MOEA1	MOEA2	MOEA3	MOEA4	MOEA/D	NSGA-II	NSGA-III
1	8.44e-02 _{8.41e-03}	9.60e-02 _{1.12e-02} +	2.43e-01 _{1.63e-02} +	1.68e-01 _{2.18e-02} +	1.76e-01 _{2.15e-02} +	3.50e-01 _{3.90e-02} +	2.75e-01 _{3.35e-02} +	3.03e-01 _{1.67e-02} +
2	8.18e-02 _{1.03e-02}	1.09e-01 _{1.25e-02} +	1.93e-01 _{1.23e-02} +	1.37e-01 _{1.05e-02} +	1.55e-01 _{1.58e-02} +	3.14e-01 _{2.26e-02} +	2.32e-01 _{1.74e-02} +	2.40e-01 _{2.24e-02} +
3	1.05e-01 _{1.24e-02}	1.57e-01 _{2.52e-02} +	2.10e-01 _{1.57e-02} +	1.43e-01 _{1.24e-02} +	2.10e-01 _{1.49e-02} +	2.84e-01 _{1.52e-02} +	1.99e-01 _{2.93e-02} +	2.23e-01 _{3.72e-02} +
4	1.43e-01 _{2.24e-02}	2.29e-01 _{2.26e-02} +	1.95e-01 _{1.98e-02} +	1.51e-01 _{1.17e-02} =	2.10e-01 _{2.46e-02} +	3.11e-01 _{2.21e-02} +	2.53e-01 _{1.93e-02} +	2.66e-01 _{1.75e-02} +
5	1.39e-01 _{2.76e-02}	1.60e-01 _{1.47e-02} +	2.05e-01 _{1.12e-02} +	1.67e-01 _{9.19e-03} +	2.11e-01 _{1.63e-02} +	2.41e-01 _{1.20e-02} +	1.68e-01 _{8.26e-03} +	1.83e-01 _{7.56e-03} +
6	1.19e-01 _{4.65e-02}	2.94e-01 _{2.04e-02} +	2.02e-01 _{1.47e-02} +	1.98e-01 _{3.21e-02} +	2.13e-01 _{8.85e-03} +	2.15e-01 _{1.94e-02} +	2.15e-01 _{2.66e-02} +	2.19e-01 _{1.67e-02} +
7	1.59e-01 _{2.55e-02}	3.71e-01 _{1.65e-02} +	1.57e-01 _{1.41e-02} =	1.88e-01 _{2.57e-02} +	2.68e-01 _{2.36e-02} +	4.74e-01 _{5.61e-02} +	2.31e-01 _{1.88e-02} +	2.65e-01 _{2.08e-02} +
8	1.22e-01 _{1.31e-02}	1.46e-01 _{1.87e-02} +	2.68e-01 _{1.70e-02} +	1.91e-01 _{1.70e-02} +	2.21e-01 _{2.17e-02} +	3.32e-01 _{1.33e-02} +	2.67e-01 _{2.31e-02} +	2.99e-01 _{1.86e-02} +
9	1.22e-01 _{1.58e-02}	1.64e-01 _{1.24e-02} +	2.40e-01 _{3.48e-02} +	1.68e-01 _{1.65e-02} +	2.24e-01 _{2.85e-02} +	3.06e-01 _{1.48e-02} +	2.56e-01 _{2.57e-02} +	2.81e-01 _{1.61e-02} +
10	1.69e-01 _{1.54e-02}	2.57e-01 _{1.43e-02} +	2.39e-01 _{4.65e-02} +	1.62e-01 _{1.33e-02} =	2.69e-01 _{2.92e-02} +	2.85e-01 _{6.33e-03} +	2.37e-01 _{1.81e-02} +	2.65e-01 _{1.45e-02} +
+/-/=		10/0/0	9/0/1	8/0/2	10/0/0	10/0/0	10/0/0	10/0/0



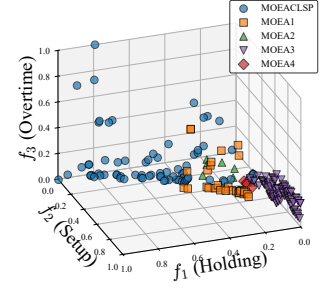
(a) Instance 1 ($T = 15$, $J = 6$)



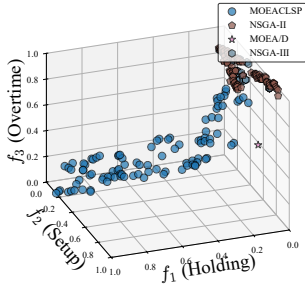
(b) Instance 4 ($T = 15$, $J = 24$)



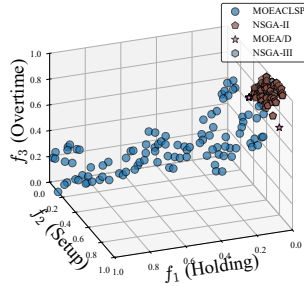
(a) Instance 2 ($T = 15$, $J = 8$)



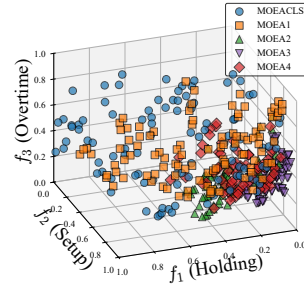
(b) Instance 5 ($T = 20$, $J = 10$)



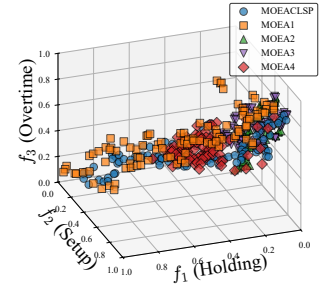
(c) Instance 7 ($T = 20$, $J = 30$)



(d) Instance 9 ($T = 30$, $J = 12$)



(c) Instance 8 ($T = 30$, $J = 6$)



(d) Instance 10 ($T = 30$, $J = 24$)

Fig. 2. Plots of the approximated PFs (best among 20 independent runs for each algorithm) obtained by MOEACLSP and other algorithms.

Fig. 3. Plots of the approximated PFs (best among 20 independent runs for each algorithm) obtained by MOEACLSP and its variants.

“+”, “-”, and “=” indicate that MOEACLSP is significantly better, worse, or not significantly different from the compared algorithm, respectively.

All experiments are conducted on a Windows 11 PC with an Intel Core i7-11700 CPU, compiled with g++ 15.2.0.

C. Comparative Results and Analysis

Tables I and III report the mean and standard deviation of HV and IGD over 20 independent runs, the top results are highlighted using different gray levels, where a darker shade indicates a better value. Besides NSGA-II, MOEA/D, and NSGA-III, four MOEACLSP variants (MOEA1-MOEA4) are also included for ablation analysis, i.e., MOEA1 removes the cluster-guided LNS, MOEA2 removes the repair procedure, MOEA3 removes the production shifting operator, and MOEA4 replaces the recombination operator with SBX and polynomial mutation.

The tables show that MOEACLSP achieves better HV and IGD on most instances, indicating better PF approximations than the compared algorithms. Only a few exceptions appear for HV on Instances 1, 2 and for IGD on Instances 7, 10, where MOEACLSP remains very close to the best results and the differences are not statistically significant.

The ablation variants help explain which components contribute most to the overall performance. MOEA2 shows the most consistent performance drop across instances, indicating that repair plays an important role in maintaining feasibility and solution quality. MOEA3 and MOEA4 also perform worse than MOEACLSP on many instances, suggesting that production shifting and the original recombination operator both support effective exploration of production plans. More notably, MOEA1 remains relatively competitive on several instances with fewer items, but becomes clearly worse on Instances 6 and 7, where the number of items is larger. This

indicates that the cluster-guided LNS is particularly effective for instances with many items, where additional neighborhood search improves both convergence and solution distribution.

Fig. 2 and 3 each show the approximated PFs on four representative instances. For each algorithm and each instance, the plotted PF corresponds to the best feasible non-dominated set among the 20 independent runs. Compared with the competing algorithms, MOEACLSP typically achieves better convergence and a more diverse distribution of solutions, while some methods miss parts of the PF or show a reduced spread in some regions. This observation is also reflected in Fig. 4, where the indicator curves show that MOEACLSP usually improves HV and IGD earlier and maintains progress longer than other algorithms. The indicator curves on Instances 6 and 7 also show that MOEA1 stagnates earlier and more noticeably, which further suggests that the cluster-guided LNS becomes more effective as the number of items increases.

VI. CONCLUSION

This paper studied a baseline three objectives MO-CLSP that follows the standard CLSP structure and considers holding, setup, and overtime costs. 1) The baseline formulation provides a reference setting for studying the MO-CLSP with holding, setup, and overtime costs. 2) MOEACLSP achieved competitive HV and IGD compared with widely used MOEAs, and additional analyses indicate the contribution of its key components. 3) The benchmark derived from standard CLSP instances enables consistent evaluation of MO-CLSP algorithms under the same instances and settings.

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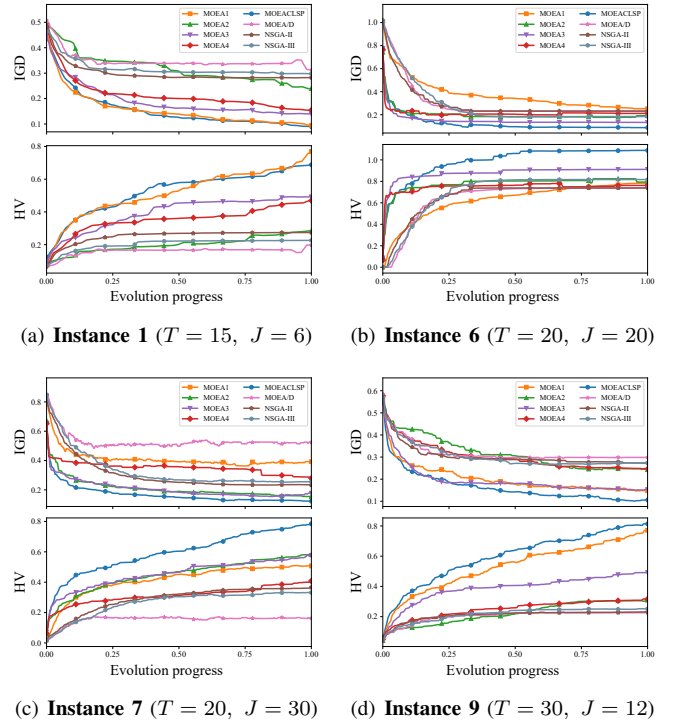


Fig. 4. Indicator curves of HV and IGD based on the best run of each algorithm.

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